AVOIDING FINITELY MANY PRIME IDEALS WHILE STAYING IN A FIXED COSET OF AN IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Prime avoidance for a single prime. Used at proof line 2826 to find y in I outside a chosen prime. Supplies the element of I used in the induction step.
- Intersection of primes with one prime not contained. Used at proof line 2827 to choose f. Provides a multiplier lying in the earlier primes but not in the current one, keeping the adjusted element out of those primes.

Lemma 0.1. *Let R be a ring. Let $x \in R$, $I \subset R$ an ideal, and $\mathfrak{p}_i$, $i = 1, \ldots, r$ be prime ideals. Suppose that $x + I \not\subset \mathfrak{p}_i$ for $i = 1, \ldots, r$. Then there exists a $y \in I$ such that $x + y \notin \mathfrak{p}_i$ for all i.*

Proof. We may assume there are no inclusions among the $\mathfrak{p}_i$. After reordering we may assume $x \notin \mathfrak{p}_i$ for $i < s$ and $x \in \mathfrak{p}_i$ for $i \geq s$. If $s = r + 1$ then we are done. If not, then we can find $y \in I$ with $y \notin \mathfrak{p}_s$. Choose $f \in \bigcap_{i<s} \mathfrak{p}_i$ with $f \notin \mathfrak{p}_s$. Then $x + fy$ is not contained in $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$. Thus we win by induction on s. □

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